

Transaction Prediction by Rebalancing the Signal

Gaussian Naive Bayes on Imbalanced Transaction Data

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Problem Framing: The Loud Pattern

- Predict future customer transaction behavior
- Binary outcome: transaction or no transaction
- Anonymized variables treated equally
- Class imbalance affected GNB model learning
- Core Problem: “How do I help the model hear the quieter signal?”

Data Cleaning: Removing Static from the Signal

- Removed null-only columns and rows
- Reduced dataset from 1,050,000 x 104 to 180,000 x 51
- Split cleaned data into training and test sets
- Balanced training data before 2nd pass with modeling

**RAW
DATA**

CLEAN

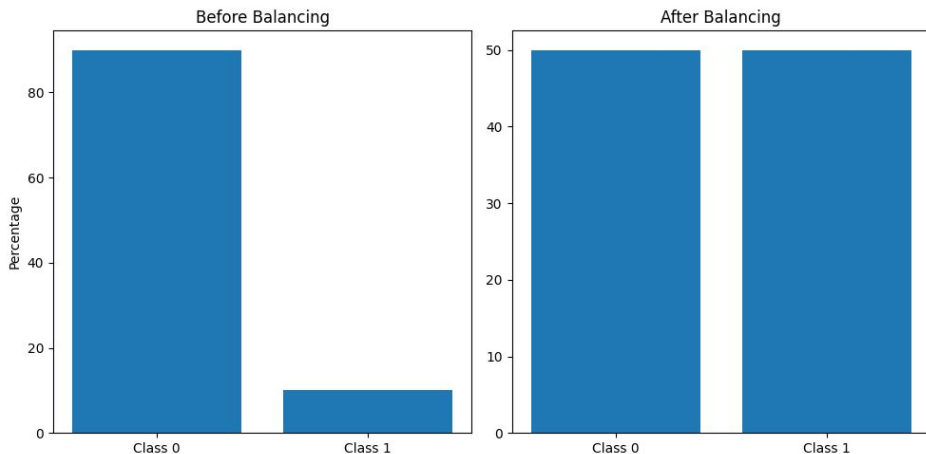
SPLIT

BALANCE

**CLEAR
SIGNAL**

Rebalancing the Signal

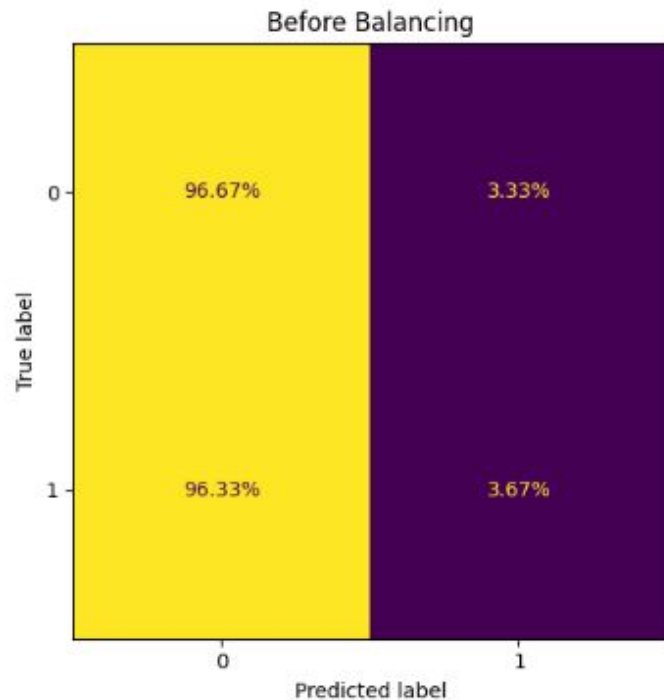
- Initial model favored the majority class
- Undersampling created a balanced training set
- Rebalancing improved minority detection



Before Balancing:

Looks Strong, Misses the Signal

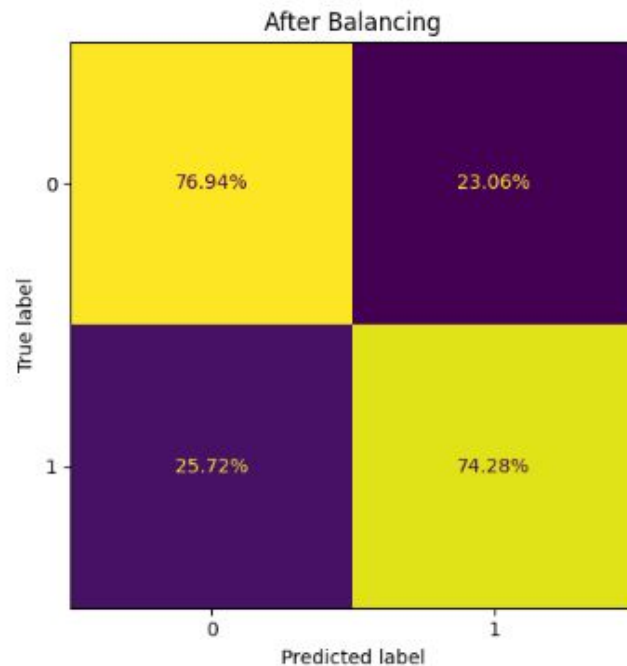
- Only **136 true positives** were identified (**3.67% recall**)
- Overall accuracy (**91%**) hid weak minority detection
- Most actual positive cases were missed
- Strong on paper, weak on signal recovery



After Balancing:

Recall =
Recovering the Real Signal

- Recall improved sharply
- True positives increased to **13,401 (74.28% recall)**
- More real positive cases were recovered
- False positives increased as tradeoff



Why the First Model Missed the Signal

- GNB estimates probability for two classes
- Early learning favored the dominant class
- Actual successful transactions were often missed

The Mix Before Rebalancing

Majority class  90%

Minority class  10%

Conclusion: The Better Mix

- Imbalanced data can hide model weakness
- Rebalancing improved minority-class detection
- Model evaluation must match the real objective
- Tradeoff between recall and false positives/ signal and noise

What is Next?

- Explore other balancing strategies
- Compare GNB with other models:
 1. logistic regression
 2. decision trees
- See whether recall can remain strong while reducing false positives
- Refine visual communication of results
- Strengthen the explanation of why balancing changed the model's behavior
- Connect evaluation metrics more clearly to the business goal

Appendix: Project Framework

- Business goal: predict future customer transaction behavior
- Machine learning type: supervised learning
- Target variable: future transaction (0 = no, 1 = yes)
- Target type: categorical
- Problem type: binary classification
- Model selected: Gaussian Naive Bayes

Because variables were anonymized, interpretation focused on model behavior rather than feature meaning.

Appendix: Data Source

Source

- CSV file stored in AWS S3 cloud storage

Dataset Size

- ~187 MB

Tools Used

- Google Colab
- Pandas

Data Dictionary

- Variables were anonymized
- No business-facing feature descriptions provided

Appendix: Cleaning Logic

- Rows and columns inspected using `.shape`
- Null values checked using `.isnull().sum()`
- Null-only columns removed
- Duplicate rows checked before modeling

Python

```
df.shape  
df.isnull().sum()  
df.dropna(axis=1, inplace=True)
```

Appendix: Exploratory Data Analysis (EDA)

- Target distribution revealed class imbalance
- Most variables were continuous numerical fields
- Histograms used for feature distribution
- Correlation heatmap used for variable relationships

Appendix: Processing + Evaluation Logic

- Train/test split applied before modeling
- Balancing introduced before final model training
- Accuracy measured overall correctness
- Recall became critical because minority cases were being missed
- Balancing improved recall before/after tradeoff
- False positives increased after balancing
- Confusion matrix showed before/after tradeoff

Appendix: Constraints

- Anonymized data
- No data dictionary, which limited feature interpretation
- Imbalanced feature distribution
- Limited scope
- Required model choice